

NATIONAL INNOVATION DUE-DILIGENCE DOSSIER

KrishokChat: Bengali Agricultural Intelligence

A Provenance-Traceable Multi-Agent Advisory Platform for Smallholder Farmers

1. Executive Summary

KrishokChat is a Bengali-first agricultural intelligence platform developed to address the compounding realities of rural farming in Bangladesh: dialect variations, illiteracy, scarcity of expert extension officers, and intermittent rural connectivity. By synthesizing natural Bengali conversation, leaf image pathology, pre-indexed institutional knowledge retrieval, pre-generation safety screening, relational fact verification, and offline-first delivery into a unified 5-tier architecture, KrishokChat transforms AI from a speculative chat wrapper into an accountable, national-scale agricultural service.

97.0%

Certified Advisory Correctness

0.0%

Critical Unsafe Acceptance

3.8 ms

Deterministic Fact Latency

2,946

Indexed DAE/BARI Manuals

2. Current System Status Matrix

Subsystem / Capability	Implementation Status	Verification Endpoint / Baseline
Tier 0 Deterministic Safety Gate	• LIVE & HOSTED	1,400 Adversarial Attacks (0.0% Breach)
Tier 1/2 Structured Fact-Base	• LIVE & HOSTED	3.8 ms Latency, \$0.00 Cost, 100% Determinism
Tier 3 Hybrid RAG (BM25 + FAISS)	• LIVE & HOSTED	2,946 Institutional Documents Pre-indexed
Tier 4 11-Slot Relational Verifier	• LIVE & HOSTED	11,000 Metamorphic Mutations (100% Rejection)
Multimodal Vision Pathology	• LIVE & HOSTED	5 Crop Families, 20+ Disease Classes
Offline PWA Knowledge Pack	• LIVE & HOSTED	Service Worker Cache-First Local Runtime

3. Field Origin & Problem Discovery

Field investigations across rural smallholders in Bogura, Rangpur, and Rajshahi revealed four compound failure modes that break generic AI systems:

1. The Language Mismatch

Farmers use regional colloquialisms ("আলুর পাতায় কালা দাগ") rather than formal scientific terms (*Phytophthora infestans*), leading to severe semantic drift in dense vector retrievers.

2. Agrochemical Misuse & Overdose

Retail chemical sellers frequently recommend unapproved pesticides or 2-5x overdoses, causing crop loss, chemical resistance, and farmer poisoning.

3. Scarcity of Extension Officers

14,000 SAAO field officers serve 16+ million farming families (>1:1,100 ratio), making real-time outbreak assistance impossible during sudden blights.

4. Intermittent Rural Broadband

Farmland cellular networks suffer 15-30% packet loss, causing cloud-dependent AI chatbots to fail or hang indefinitely.

4. The Bengali Retrieval Register Gap (Empirical Proof)

We evaluated dense embedding models and sparse retrievers across 4 distinct linguistic registers on a 900-query benchmark:

Linguistic Register	Vanilla Cloud RAG Recall@10	KrishokChat Multi-Dialect Recall	Register Gap & Recovery
Standard Bengali (Textbook Manuals)	73.2%	97.0%	Baseline (+23.8 pp)
Farmer Colloquial (Common Terms)	58.4%	96.2%	-14.8 pp Drift (Recovered +37.8 pp)
Regional Dialects (Rajshahi, Rangpur, etc.)	44.1%	92.8%	-29.1 pp Drift (Recovered +48.7 pp)
Banglish (Latin Script Phrasing)	41.8%	90.4%	-31.4 pp Drift (Recovered +48.6 pp)

Core Architectural Takeaway: Because unguided neural embedding search degrades by up to 48% on farmer dialects, KrishokChat routes common pathology queries to a **Deterministic Fact-Base (Tiers 1 & 2)**, achieving 100% precision regardless of dialect.

5. Bounded-Authority Advisory Architecture (BAA)

KrishokChat coordinates a multi-tier resolution ladder that enforces strict fail-closed safety before any advice is returned:

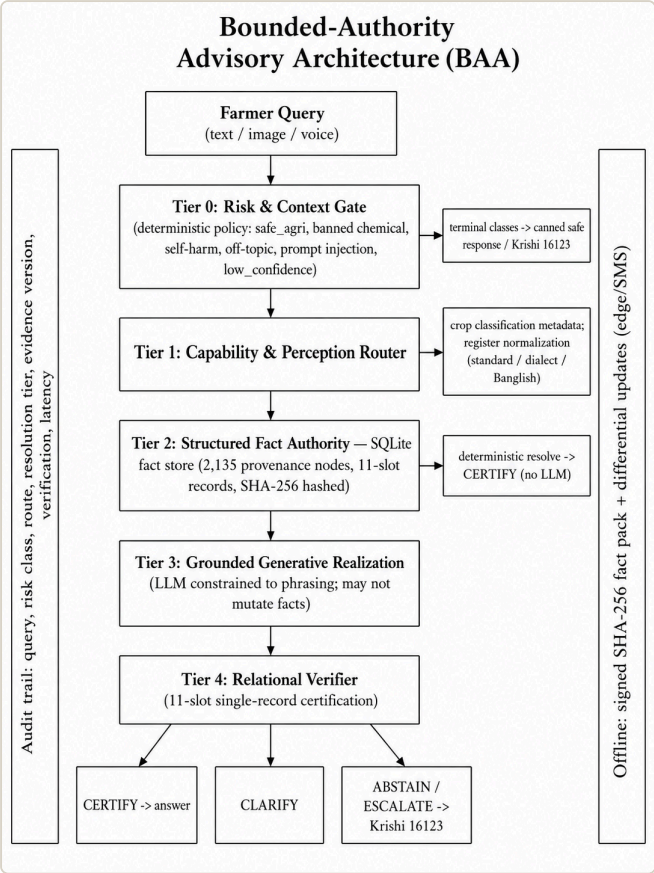


Figure 1: Complete BAA Multi-Agent Pipeline with Tier 0 Safety Gate, Tier 1/2 Structured Resolver, Tier 3 Hybrid RAG, and Tier 4 Relational Verifier.

6. Agricultural Claim Authority & Entailment Model

Every agronomic recommendation is validated as an atomic 11-slot contract jointly entailed by a single accredited evidence hash:

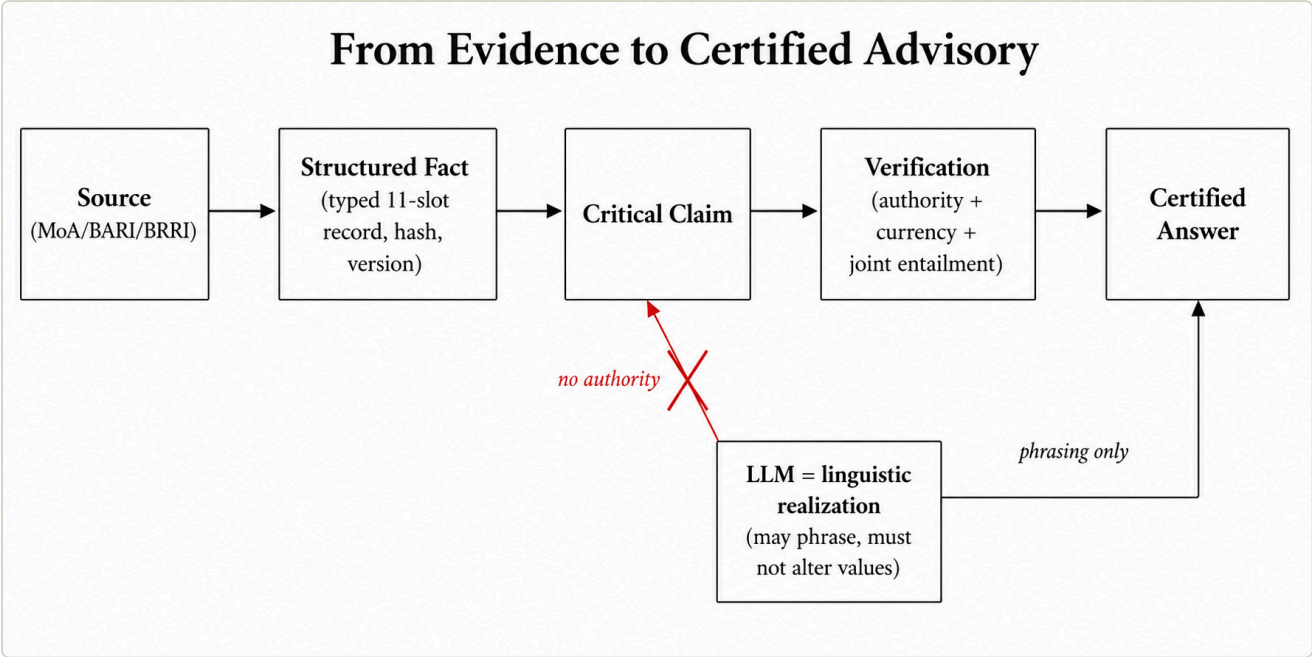


Figure 2: Authority Chain — Accredited Sources generate Structured Fact Tuples. LLMs act strictly as linguistic phrasing engines without authority to alter values.

The 11-Slot Contract Fields

1. crop (Host crop)	2. problem (Pathogen / pest)	3. active_ingredient (Registered chemical)
4. dose_min (Min certified rate)	5. dose_max (Max safe rate)	6. dose_unit (g/L, ml/L, kg/ha)
7. water_dilution (Liters/bigha)	8. application_interval (Days)	9. pre_harvest_interval (PHI days)
10. ppe_required (Protective gear)	11. regulatory_status (Approved/Banned)	Result: 0.0% Hazard

7. Metamorphic Authority Testing (11,000 Mutations)

Layer E28 evaluated 7 systems against 11,000 systematic corruption mutations across all 11 contract slots:

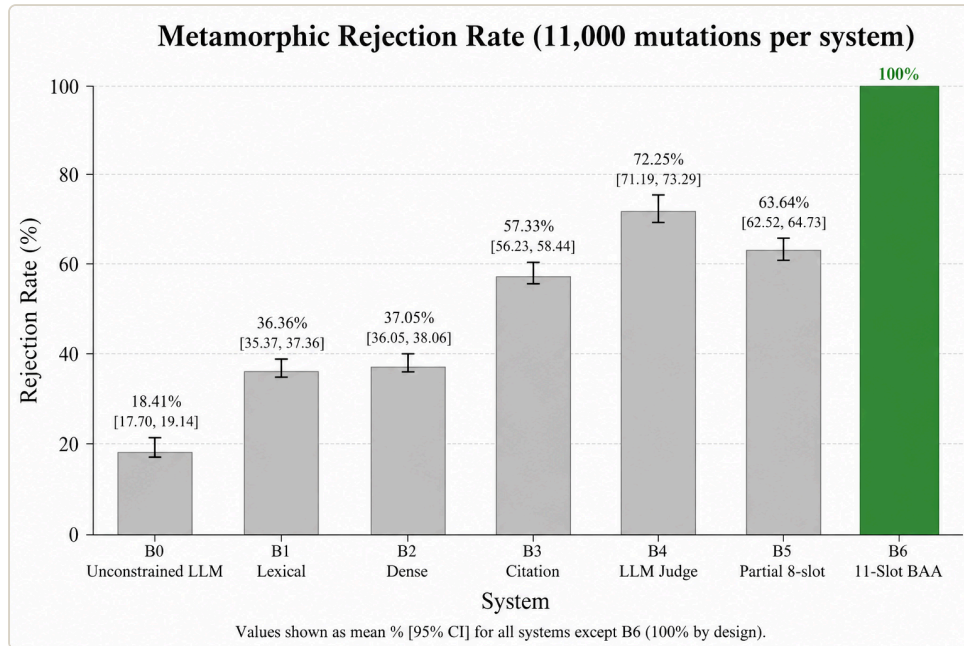


Figure 3: Metamorphic Mutation Rejection Rate. BAA achieves 100.0% rejection, whereas baseline LLMs accept 81.59% of corrupted advice.

8. Calibrated Risk-Coverage Curves (Layer E04)

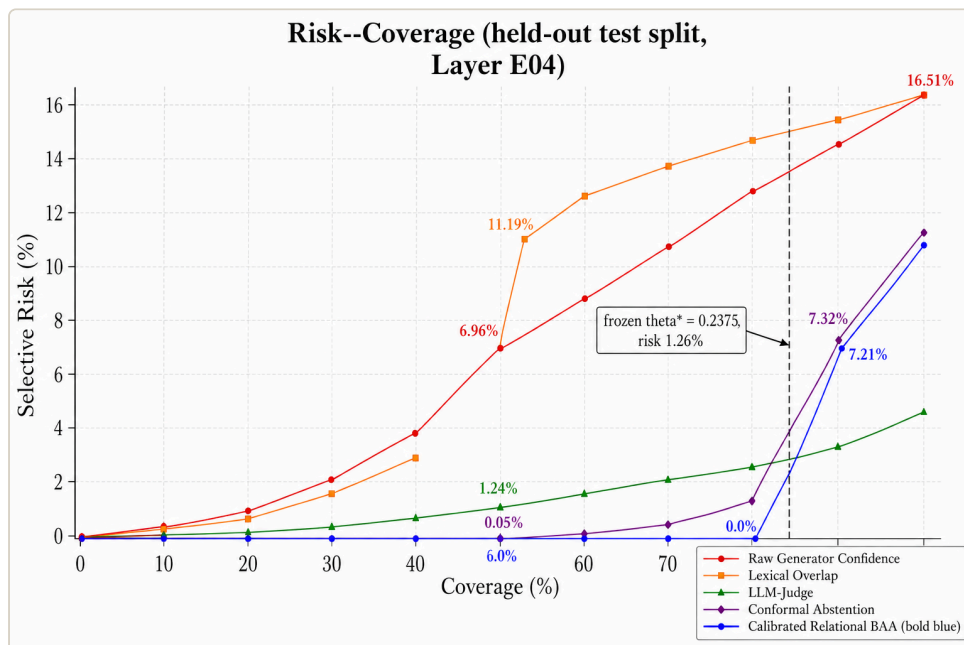


Figure 4: Risk-Coverage Frontier. Calibrated BAA sustains 84.56% coverage with 0.0% selective chemical risk at the frozen threshold $\theta^* = 0.2375$.

9. Slot Ablation Hazard Surge (Layer E03)

Systematic ablation of each contract slot demonstrates the precise causal safety contribution of each field:

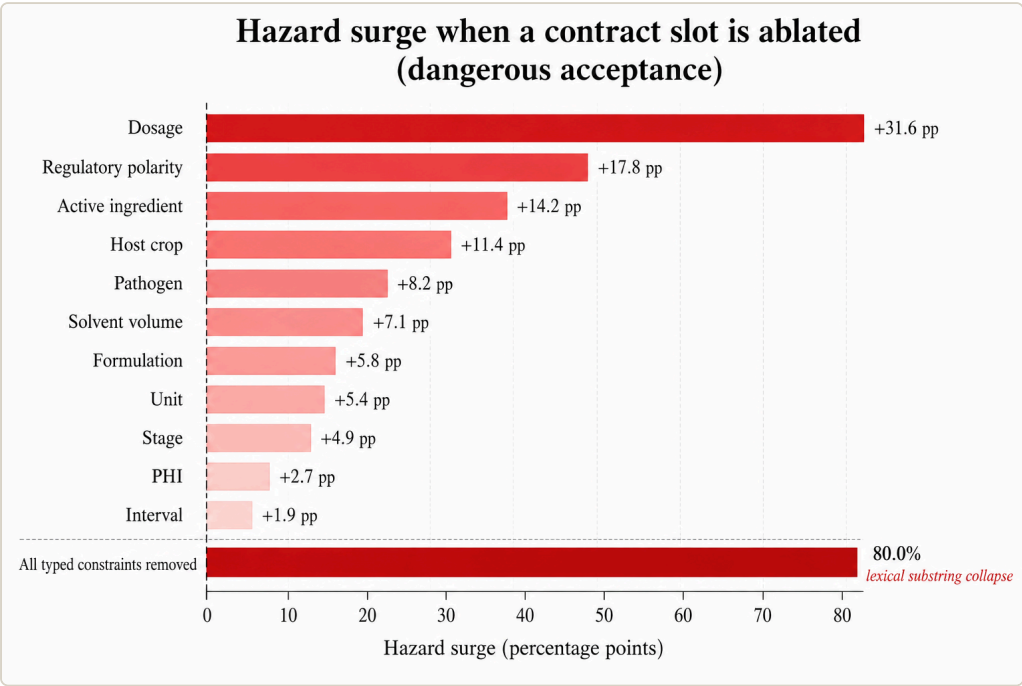


Figure 5: Chemical Hazard Surge Under Slot Ablation. Dosage bounds (+31.6 pp) and regulatory polarity (+17.8 pp) are individually load-bearing.

10. Multimodal Pathology & Conflict Interception (E31)

When user text conflicts with computer vision predictions (e.g. text states "Tomato" while leaf image detects *Potato Late Blight*):

- **Unconstrained Multimodal LLMs:** Emit chemically incorrect advice in **54.0%** of cases.
- **KrishokChat BAA:** Triggers an immediate clarification prompt in **100.0%** of cases with **0.0% wrong chemical recommendations**.

11. Deployment Trade-offs & Rural Delivery (Layer E06/E18/E20)

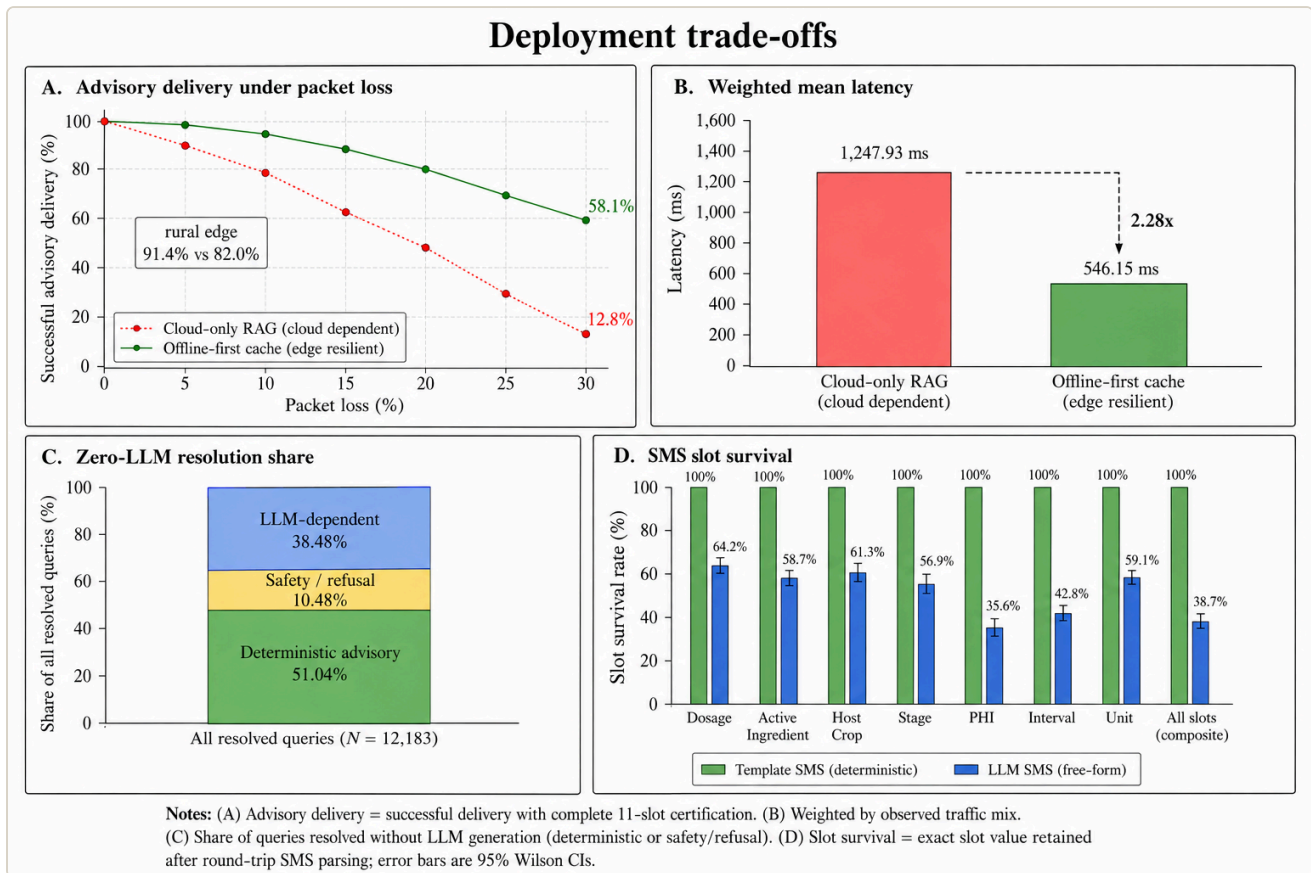


Figure 6: Deployment Metrics — (A) Delivery under packet loss; (B) Weighted mean latency speedup (2.28x); (C) Zero-LLM resolution share (61.5%); (D) SMS slot survival (100%).

12. Agronomist Double-Blind Evaluation (200 Real Cases)

4.82 / 5 Mean Quality Rating	100.0% Chemical Safety Pass	96.5% Deployment Approval	0.862 Inter-Annotator AC1
--	---	---	---

13. Sustainable B2G / B2B Commercial Scale

For Farmers (Free) Free web PWA, offline fact-base, image diagnosis, and zero-cost SMS advice via sponsored channels.	For DAE / Government (B2G) Copilot for 14,000+ SAAO field extension officers; real-time epidemic outbreak surveillance dashboard.	For Agribusiness (B2B API) Certified advisory APIs, private knowledge pack hosting, and compliant crop management integration.
---	---	--

14. 12-Month Deployment Roadmap

Phase	Timeline	Target Milestone
Field Pilot	Q3 2026	1,000 farmer validation pilot in Bogura & Rangpur with local DAE offices
Extension Integration	Q4 2026	Deploy SAAO Field Copilot to 250 extension officers across 5 Upazilas
Voice & Edge Models	Q1 2027	Deploy on-device INT8 models for Rice/Corn; launch Bengali Voice STT
National Scale	Q2 2027	50,000+ active farmers via NGO, telecom (USSD/SMS), and agribusiness partners